

Q1 - 24 January - Shift 1

Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$; Then at $x = 0$

Space for your notes:

- (1) f is continuous but not differentiable
- (2) f is continuous but f' is not continuous
- (3) f and f' both are continuous
- (4) f' is continuous but not differentiable

Q2 - 25 January - Shift 2

If the function

$$f(x) = \begin{cases} (1 + |\cos x|) \frac{\lambda}{|\cos x|}, & 0 < x < \frac{\pi}{2} \\ \mu, & x = \frac{\pi}{2} \\ e^{\frac{\cot 6x}{\cot 4x}}, & \frac{\pi}{2} < x < \pi \end{cases}$$

Space for your notes:

is continuous at $x = \frac{\pi}{2}$, then

$9\lambda + 6 \log_e \mu + \mu^6 - e^{6\lambda}$ is equal to

- (1) 11
- (2) 8
- (3) $2e^4 + 8$
- (4) 10

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Answer Key

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(As per Official NTA Key released on 2 Feb)

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Q1 (2)

Q2 (4)

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#MathBoleTohMathonGo

Q1 (2)

Continuity of $f(x)$: $f(0^+) = h^2 \cdot \sin \frac{1}{h} = 0$

$f(0^-) = (-h)^2 \cdot \sin \left(\frac{-1}{h} \right) = 0$

$f(0) = 0$

$f(x)$ is continuous

$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \frac{h^2 \cdot \sin \left(\frac{1}{h} \right) - 0}{h} = 0$

$f'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \frac{h^2 \cdot \sin \left(\frac{-1}{h} \right) - 0}{-h} = 0$

$f(x)$ is differentiable.

$f'(x) = 2x \cdot \sin \left(\frac{1}{x} \right) + x^2 \cdot \cos \left(\frac{1}{x} \right) \cdot \frac{-1}{x^2}$

$$f'(x) = \begin{cases} 2x \cdot \sin \left(\frac{1}{x} \right) - \cos \left(\frac{1}{x} \right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$\Rightarrow f'(x)$ is not continuous (as $\cos \left(\frac{1}{x} \right)$ is highly

Q2 (4)

$$\Rightarrow \lim_{x \rightarrow \frac{\pi^+}{2}} e^{\frac{\cot 6x}{\cot 4x}} = \lim_{x \rightarrow \frac{\pi^+}{2}} e^{\frac{\sin 4x \cdot \cos 6x}{\sin 6x \cdot \cos 4x}} = e^{2/3}$$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi^-}{2}} (1 + |\cos x|)^{\frac{\lambda}{|\cos x|}} = e^{\lambda}$$

$$\Rightarrow f(\pi/2) = \mu$$

For continuous function $\Rightarrow e^{2/3} = e^{\lambda} = \mu$

$$\lambda = \frac{2}{3}, \mu = e^{2/3}$$

Now, $9\lambda + 6\log_e \mu + \mu^6 - e^{6\lambda} = 10$